

→ What is Regression?

Regression is a method used to find a relationship b/w two (or more) things.

and use that relationship to make predictions.

→ Regression means to show the relationship b/w the variables

Note:- regression means to show the relationship b/w the variables and can i say these variables are features of data sets like name column, id, etc.

Answer: Yes it is correct that regression means to show the relationship b/w variables.

Bot:

Regression shows the relationship and uses it to predict values

so a bad feature like name, id "x"

Note - all columns (features) are useful for regression!

example of bad features:

- Name

- id

- Rollno

These are just labels/identifiers, they don't affect the outcomes.
ex: Student name = "Ali"
does not affect marks.

• $Id = 123 \rightarrow$ meaningless for prediction!
so Regression does not use these.

what kind of features Does regression use?

- Regression uses meaningful variables that influences the result.

$\rightarrow$ Good example!

Predict marks using!

- study hours.
- sleep time.
- Attendance.

so

Regression is a statistical method used to find the relationship b/w variables and predict a dependent variable using one or more independent variables

7 Types of Regression!

1. Linear Regression!

- 1. is a method used to
- 1. find a straight-line relationship b/w variables.
- 2. and predict values.

- Why does linear regression find a straight line!

That means:

when x increases, y changes at a constant rate.

and simplest way to represent a constant relationship is.

$$y = a + bx.$$

A straight line naturally shows.

- constant increase
- constant decrease

ex!

- every extra hour of study $\rightarrow +10$ marks
- Every 1°C increase $\rightarrow +5$ sales.

The change is uniform (same every time).

why not curves?

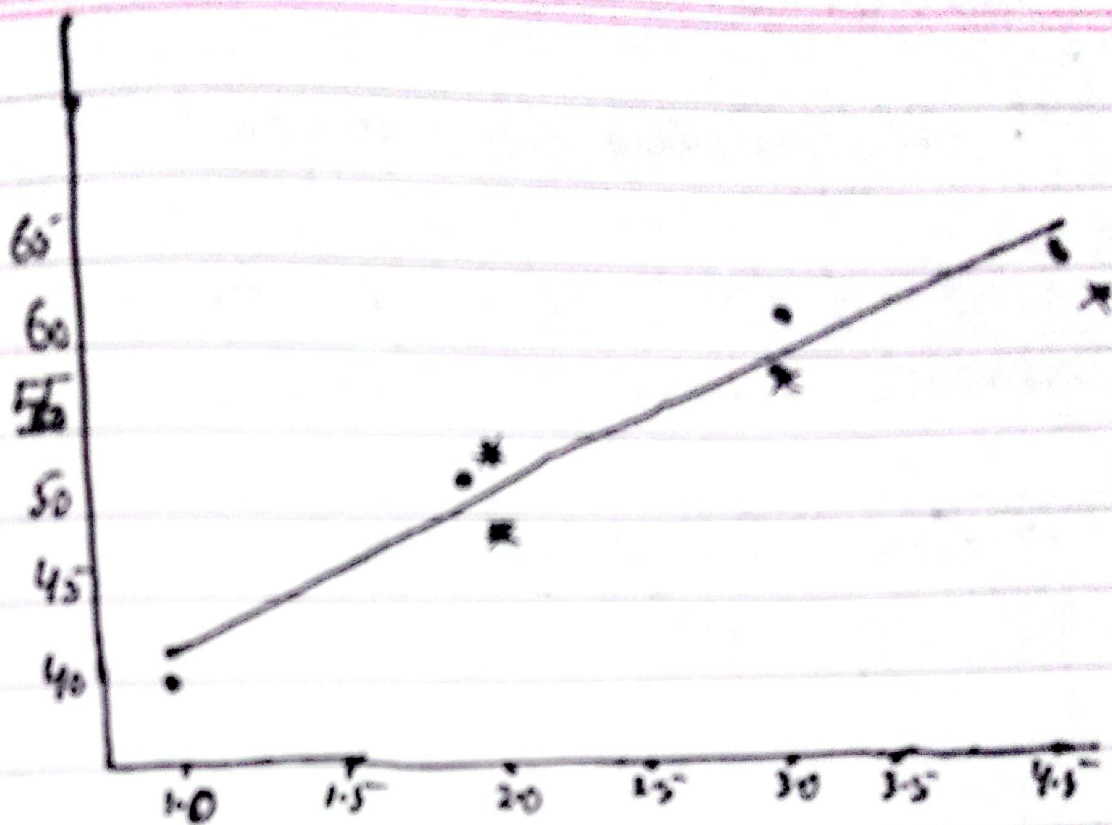
you can curves $\rightarrow$ but then it's not linear regression anymore.

Linear regression is specially designed for.

1. simple relationships.
2. easy interpretation.
3. quick prediction.

$\rightarrow$ curves are used in

1. polynomial reg.
2. non-linear models



Step 1, simple data set.

Study hours (x)	Marks (y)
1	40
2	50
3	60
4	65

Yeh perfect line nh hai (best line
 Hoda off hai.
 is liye Linear reg ki zarorat padti hai)

Step 2:

Understanding the graph!

1 Dots - real Data points.

2 line - best fit line.

→ line has point se exactly
nhi guzarti.

→ lekin sab ke beech ka best
balance banati hai.

Step 3: Best fit line kya hoti hai?

Best fit line =

aisi line jo sab points ke
sab se kareeb ho overall

Na sirf ek point ke pass!

Balki sab ke liye balanced.

step 4: Error kya hota hai?

$$\text{Error} = (\text{actual value} - \text{predicted value})$$

ex:-

Agar line predict kare

$$x = 4 \rightarrow$$

$$y = 67$$

lekin actual hai

$$y = 65$$

$$\text{error} = 2.$$

step 5: line kaise choose hoti hai?

1. Linear regression random line nahi banata.

2. Yeh choose karta hai woh line jo

total error ko minimize kare.

specially

errors ko square karke add karta hai. (isse khte hai least square)

steps

simple intuition of minimizing error.

socho 3 lines hain!

1 line A $\rightarrow$ sabi se upar

2 line B $\rightarrow$ beech mein.

3 line C $\rightarrow$ neeche

line B usually best hoti hai kyun ki!

- upar wali points ka error kam

- neeche wale points ka error kam.

Balance = minimum total error.

step 7

Best fit line ka milb hai

Given data kai hisab se,
ye responsible prediction hai.

step 8: Final understanding:

1. points real data.
2. line = prediction model.
3. Distance from line = error.
4. Best fit line = smallest total error.

Note: Agar line points kai
kareeb ho to!

Agar best fit line data points
ke bohot kareeb hai.

1. error chota hoga.

2. model achha fit hai

3. prediction zyada accurate hongi.